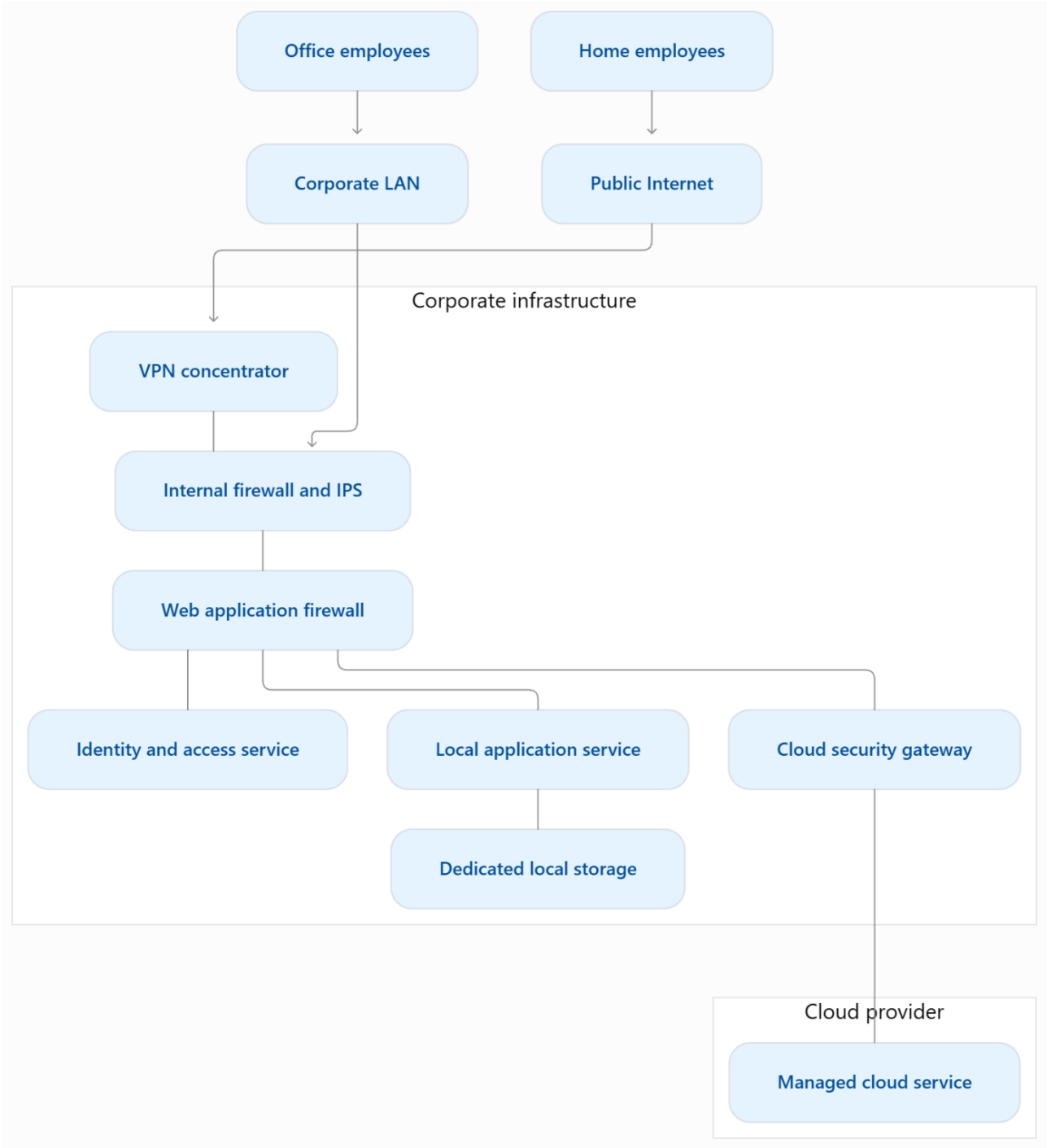
**Performance evaluation of a hybrid corporate intranet**

Your task is to model a corporate intranet, evaluate its performance and recommend an investment under a changing workload.

**1. System architecture**

Employees access the intranet either from the office or from home. The company operates some applications on premises and uses a cloud provider for other services.

The diagram shows infrastructure connectivity and dependencies. It does **not** specify queueing stations or job routing. Use the operational description below to construct your model.




Connections represent access or service dependencies. Response traffic is omitted.

**2. Workload and request processing**

During the baseline observation period:

- Office employees generate a stream of **20 transactions/s**.
- Home employees generate an independent Poisson stream of **10 transactions/s**.
    - Hint: both requests can be joined in a single stream, that routes employees requests with probability *2/3*, and home employees with probability *1/3*.
- These rates are externally imposed and do not decrease when response times increase. There is no fixed population of outstanding transactions.
- Each transaction independently requires a local service with probability **0.70**, or a cloud service with probability **0.30**, regardless of employee location.

A home transaction incurs an additional **40 ms of network latency** and requires one operation at the VPN concentrator. An office transaction requires neither.

Every transaction then undergoes, in order:

1. One firewall and intrusion-prevention inspection.
2. One identity and access check.
3. One web application firewall inspection.

The identity check validates an existing session token and its permissions. Interactive login and MFA are outside the measured transaction.

For a **local transaction**:

- The local application performs one processing operation.
- The transaction then requires at least one storage operation.
- After each storage operation, another storage operation is required with probability **0.50**; otherwise the transaction completes.
- The application releases its processing resource before storage access begins. It performs no further processing after storage access. The application server thus becomes immediately available to process a new request.

For a **cloud transaction**:

- The cloud security gateway performs one operation, covering outbound policy enforcement and data inspection.
- The transaction incurs **60 ms of cloud network latency**.
- The managed cloud service performs one processing operation, after which the transaction completes.

- Cloud-provider storage is included in the managed-service measurement and is not separately identifiable.

All transactions succeed. There are no retries, timeouts, cache hits or background workloads.

**3. Resource measurements and assumptions**

Each processing resource below handles **one operation at a time**, in arrival order, with unlimited waiting space. Service times are independent and exponentially distributed, with the listed means. A resource is shared by all transactions that use it.

| **Resource** | **Mean service time per operation** |
|---|---|
| VPN concentrator | 40 ms |
| Internal firewall and IPS | 3 ms |
| Identity and access service | 4 ms |
| Web application firewall | 6 ms |
| Local application service | 12 ms |
| Dedicated local storage | 18 ms |
| Cloud security gateway | 8 ms |
| Managed cloud service | 25 ms |

Network latency does not consume any of these processing resources. Arbitrarily many transactions may experience network latency simultaneously.

The 40 ms home-network latency and 60 ms cloud-network latency are **total additional delays per transaction**, including both directions: moreover, these delays are not shown in the proposed picture. Each applicable delay must be counted once.

Processing measurements already include the relevant request and response processing work. Do not add another security operation for returning traffic. Other LAN, transmission and client-side costs are negligible.

Security logging is asynchronous and assumed to have no measurable effect on these resources.

Response time is measured from transaction submission to completion, including all applicable network delays and waiting times.

## 4. Tasks

### A. Construct and explain your model

Build a JMT model that matches the proposed system.

Explain briefly:

- Why your workload representation matches the description.
- How you represent processing capacity and network latency.
- How you represent repeated storage operations.

Submit a screenshot of the model and the native JMT file.

### B. Establish and validate the baseline

Run the baseline configuration. Report:

- Mean response time for the transactions.
- Total completion throughput.
- Utilization of every processing resource.

Identify the resource that would saturate first if both arrival rates increased proportionally. Estimate the corresponding total arrival rate.

### C. Evaluate a change in working patterns

The company changes its working policy. Total arrival rate remains **30 transactions/s**, but **75% of transactions now originate from home**. The local/cloud service mix is unchanged.

Before running the new configuration, record:

- Which resource utilizations you expect to change.
- Which you expect to remain unchanged.
- Whether office response time should change.

Then simulate the new workload and compare results with your predictions.

Management claims:

“The number of transactions per second has not changed, so performance should remain essentially unchanged.”

Evaluate this claim using your model.

**D. Choose and defend an upgrade**

Under the workload in Task C, the company can fund exactly one of two equally priced upgrades:

| Upgrade | Change |
|---|---|
| S — Faster storage | Mean storage service time decreases from 18 ms to 9 ms |
| V — Faster VPN processing | Mean VPN service time decreases from 40 ms to 20 ms |

Neither upgrade changes network latency, routing probabilities, security functionality or any other service time.

Simulate both alternatives separately, starting from the Task C configuration.

The operational target is a **mean home-transaction response time below 300 ms**.

Recommend an upgrade using:

1. The estimated home response time.
2. The resource that would saturate first after the upgrade if arrival rates increased proportionally, keeping the 75% home share.

Finally, assess this statement:

“If we eliminate the 60 ms cloud-network latency, cloud-gateway utilization must decrease.”

Give a numerical prediction for cloud-gateway utilization and explain whether your conclusion depends on the workload assumption.

**5. Submission**

Submit:

- The native JMT model for each of the four configurations (initial, new workload, alternative S and alternative V).
- One compact results table covering all configurations, creating using any tool of your choice (i.e. Word, Excel, PowerPoint, Latex…) in PDF.

Preserve your initial predictions even if the simulation contradicts them. Explain any corrections. Generic answers without evidence from your model receive limited credit.

You may consult your notes, search the web and use AI tools. You are responsible for checking all assumptions, model settings and conclusions. Answers must refer to your submitted JMT model and numerical results.

Enter your answers in the following form:

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https://forms.cloud.microsoft/Pages/ResponsePage.aspx?id=K3EXCvNtXUKAjjCd8ope67-7CBR7gDJEgHF_krAEqPhUMEpFQ0RHRVg4WTRIREpZTlM2OUtZOTVGMS4u